

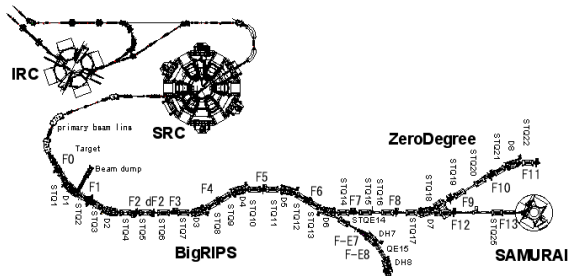
Why SAMURAI73 Needs Two Polarimeters

Full tensor reconstruction and locating p_{yy} loss

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IVR needs the tensor state delivered to the target



First IVR run

Start with vertical p_{yy} ;
longitudinal p_{zz} is harder
to prepare.

After SRC

What polarization leaves the accelerator chain?

After BigRIPS

What polarization reaches SAMURAI73?

Two stations are needed to determine all degrees of freedom of the spin-density matrix.

SAMURAI73 two-station polarimetry

Spin-1: 8 general degrees, 5 tensor, 1 prepared

For $w_a \geq 0$ and $\sum_a w_a = 1$,

$$\rho = \sum_a w_a |\psi_a\rangle \langle \psi_a|.$$

A normalized 3×3 Hermitian ρ has

$$8 = 3_{\text{vector}} + 5_{\text{tensor}}$$

independent real parameters.

An axial tensor prepared about $\hat{n}$ is

$$\mathbf{P}_0^{(2)}(q, \hat{n}) = \frac{q}{2} (3\hat{n}\hat{n}^T - I).$$

$$\hat{n} = \hat{y} : \quad \mathbf{P}_0^{(2)} = \frac{p_{yy}}{2} \text{diag}(-1, 2, -1),$$

$$\hat{n} = \hat{z} : \quad \mathbf{P}_0^{(2)} = \frac{p_{zz}}{2} \text{diag}(-1, -1, 2).$$

Tensor only: 5 degrees. Fixed preparation axis: one amplitude q .

Linear BMT preserves vector and tensor ranks

1. Magnetic field rotates the spin in the lab

$$\rho_{\text{lab},f} = U_s \rho_0 U_s^\dagger, \quad R_s \in SO(3).$$

Thomas–BMT determines R_s along the trajectory.

2. The downstream beam defines new axes

$$\rho_{b,f} = U_o^\dagger \rho_{\text{lab},f} U_o.$$

R_o follows the transported momentum and defines the local x_b, y_b, z_b frame.

$$\boxed{R_{\text{rel}} = R_o^\text{T} R_s}, \quad \boldsymbol{p}_{b,f} = R_{\text{rel}} \boldsymbol{p}_0, \quad \boldsymbol{P}_{b,f}^{(2)} = R_{\text{rel}} \boldsymbol{P}_0^{(2)} R_{\text{rel}}^\text{T}.$$

Lowest order: a common rotation moves the polarization axis, but vector and tensor ranks remain separate.

BMT field spread cannot convert tensor to vector

Each phase-space point $\xi = (x, x', y, y', \Delta p/p, \dots)$ has its own calculated rotation:

$$\mathbf{p}_f(\xi) = R_\xi \mathbf{p}_0, \quad \mathbf{P}_f^{(2)}(\xi) = R_\xi \mathbf{P}_0^{(2)} R_\xi^T.$$

The polarimeter measures the ensemble average:

$$\bar{\mathbf{p}}_f = \int d\xi f(\xi) R_\xi \mathbf{p}_0, \quad \overline{\mathbf{P}^{(2)}}_f = \int d\xi f(\xi) R_\xi \mathbf{P}_0^{(2)} R_\xi^T.$$

If $\mathbf{p}_0 = 0$, then exactly

$$\bar{\mathbf{p}}_f = 0.$$

The averaged tensor can become biaxial and require all

5 tensor components.

Different $B(\xi)$ under linear BMT can open up to 5 tensor degrees; it cannot give tensor→vector conversion.

Rank mixing opens all 8 state parameters

Tensor–vector conversion requires spin dynamics beyond the linear BMT term. For example,

$$H_{\text{spin}} = -\mu \mathbf{S} \cdot \mathbf{B} - \beta_T (\mathbf{S} \cdot \mathbf{B})^2 + \dots, \quad \bar{\rho}_f = \int d\xi f(\xi) \mathcal{E}_\xi[\rho_0].$$

Coherent rank mixing

Tensor magnetic/electric polarizability or other quadratic spin couplings can generate

$$\mathbf{p}_0 = 0, \mathbf{P}_0^{(2)} \neq 0 \quad \longrightarrow \quad \mathbf{p}_f \neq 0.$$

Open or selective transport

Spin-dependent loss, material interactions, collimation, and acceptance can change both ranks and the density-matrix eigenvalues.

Once rank mixing is allowed, the fit must allow the full $8 = 3_{\text{vector}} + 5_{\text{tensor}}$ parameters.

Two stations can recover all 8 degrees

Nominal p_{yy} monitoring

$$\frac{N_L + N_R - N_U - N_D}{N_L + N_R + N_U + N_D} \Rightarrow p_{yy}.$$

One station measures the amplitude if the beam is assumed to remain a pure p_{yy} state.

General-state test

One L/R/U/D station is blind to tensor p_{xy} and vector p_z .

A complementary second view rotates these directions into measured combinations.

Calibrated multi-angle measurement	tensor state	full density matrix
one station	4/5	6/8
two stations with a characterized complementary transfer map	5/5	8/8

Full ρ identifies what changed and where

A lower p_{yy} component has three distinct interpretations. Use

$$Q_T = \sqrt{\frac{2}{3} \text{Tr}[(\mathbf{P}^{(2)})^2]}, \quad V = |\mathbf{p}|, \quad \Pi = \text{Tr}(\rho^2).$$

measured change	physical interpretation
Q_T unchanged, $V = 0$, tensor axis changes	ordinary coherent rotation
Q_T changes, V appears, Π unchanged	coherent tensor–vector conversion
Π decreases	incoherent ensemble depolarization

Changed already after SRC $\Rightarrow$ upstream. Changed only after BigRIPS $\Rightarrow$ between the stations.

Who is the contact for an after-SRC polarimeter?

Could a polarimeter be installed just downstream of SRC?

Who should we contact to discuss

- ▶ the available beam-line location and longitudinal space;
- ▶ vacuum, mechanical, electrical, and detector interfaces;
- ▶ beam optics, operation, radiation safety, and installation constraints?

Requested next step: identify the responsible beam-line contact and arrange an interface discussion.